

Design Document using SA/SD Methodology for Automated To-Do List Application

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1 Data Flow Diagram (DFD) Model

1.1 Level 0 DFD

1.1.1 Context Diagram

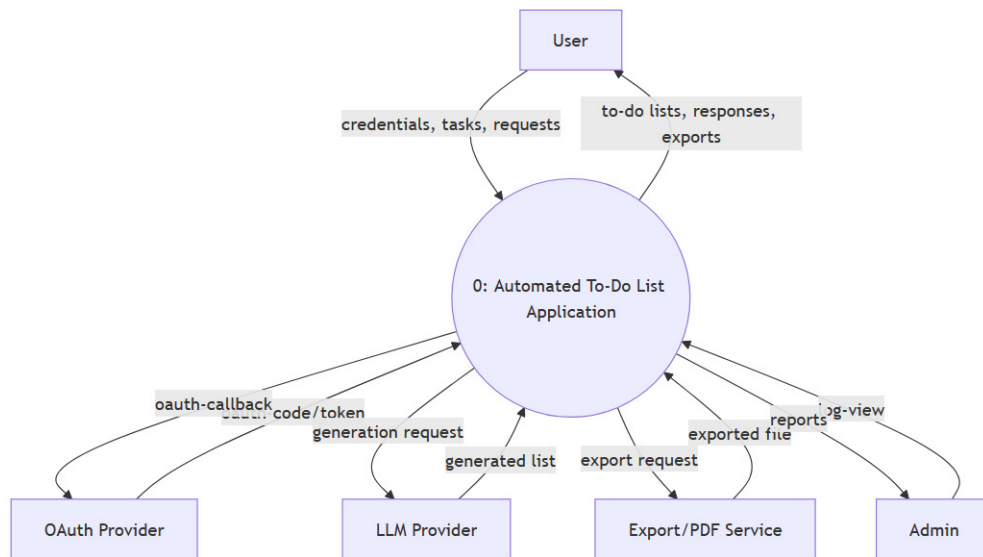


Figure 1: Level 0 Context Diagram

1.2 Level 1 DFD

1.2.1 Top-Level Functional Decomposition

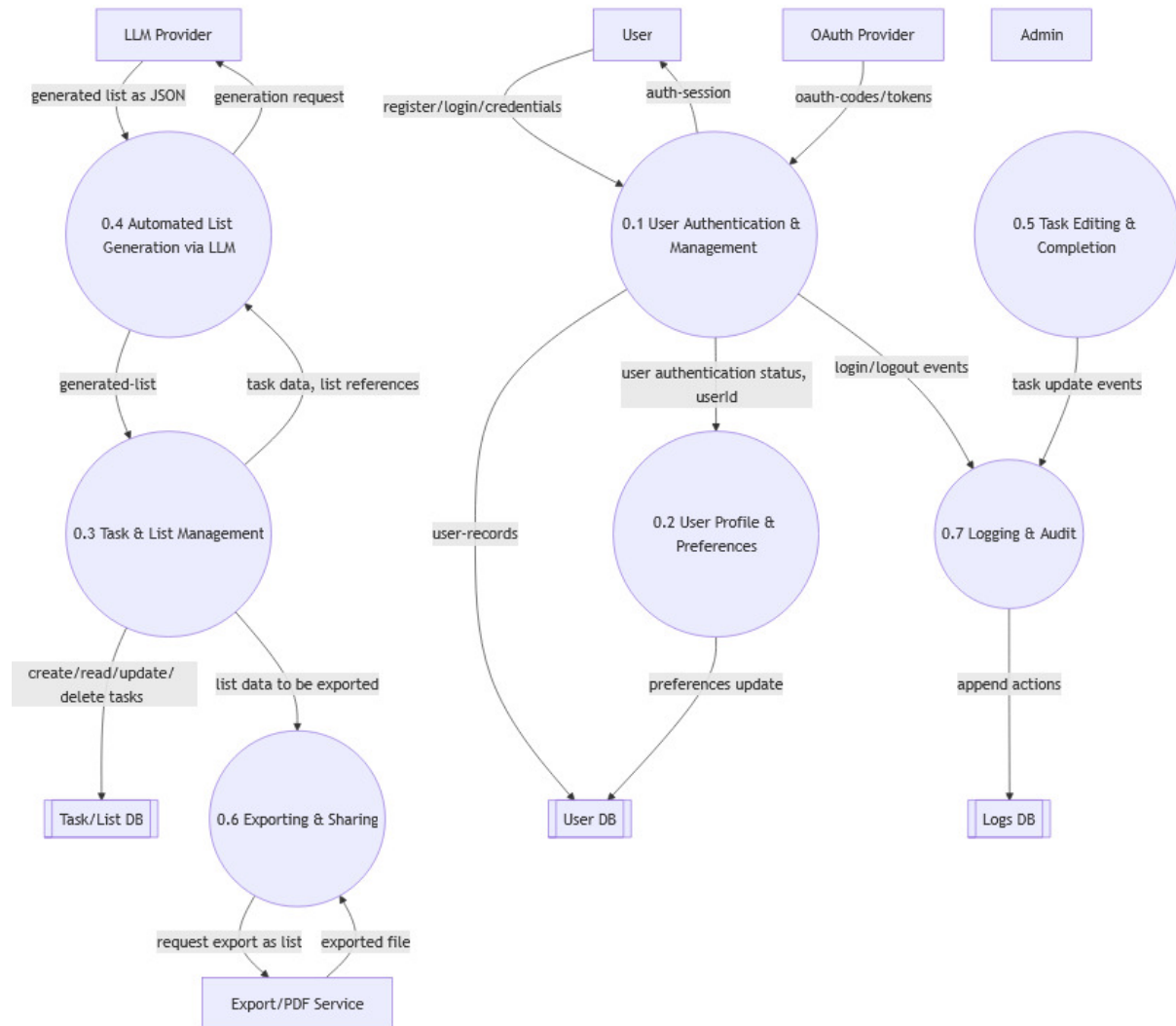


Figure 2: Level 1 DFD with 7 functional groups

1.3 Level 2 DFDs

1.3.1 0.1 User Authentication & Management

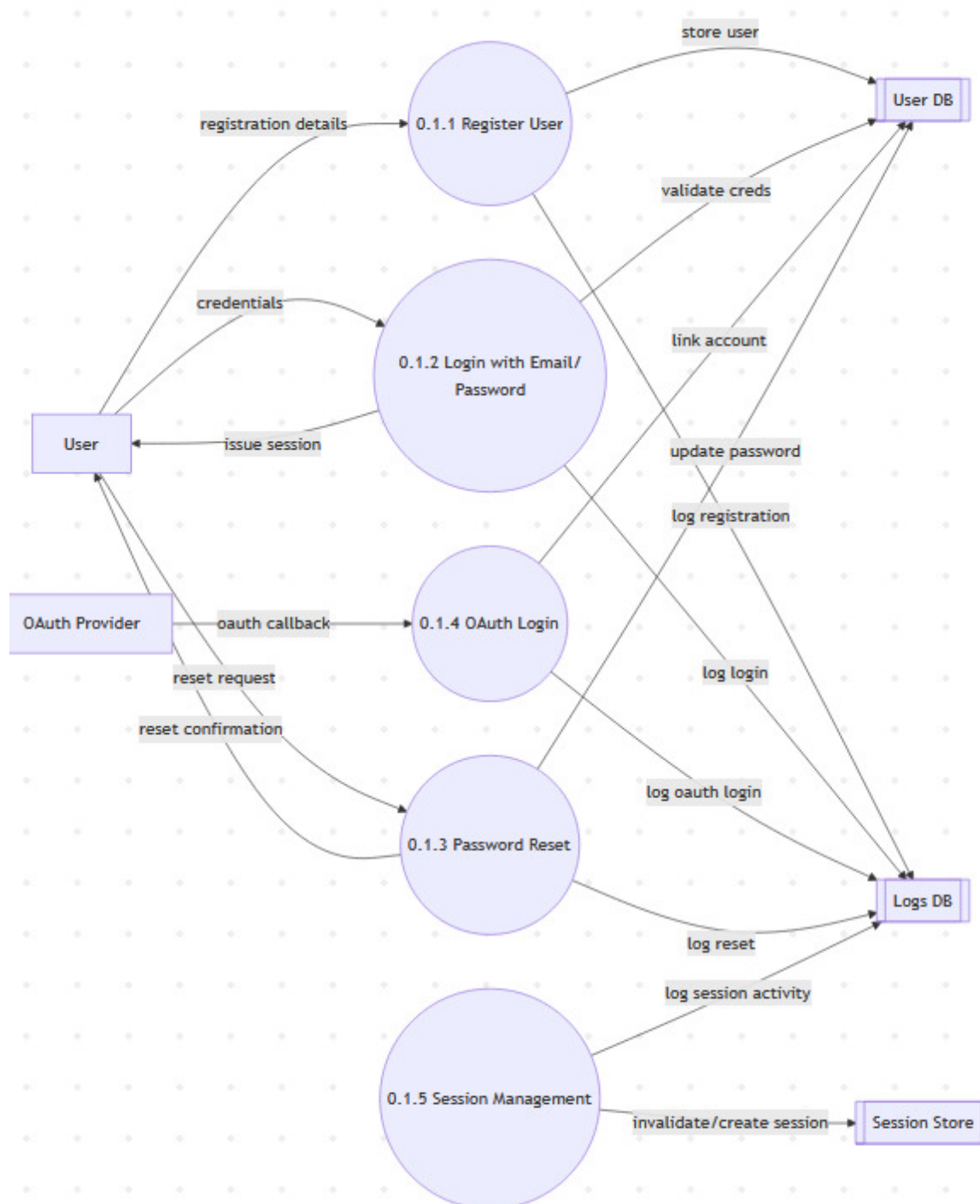


Figure 3: Level 2 DFD for 0.1 User Authentication & Management

1.3.2 0.2 User Profile & Preferences

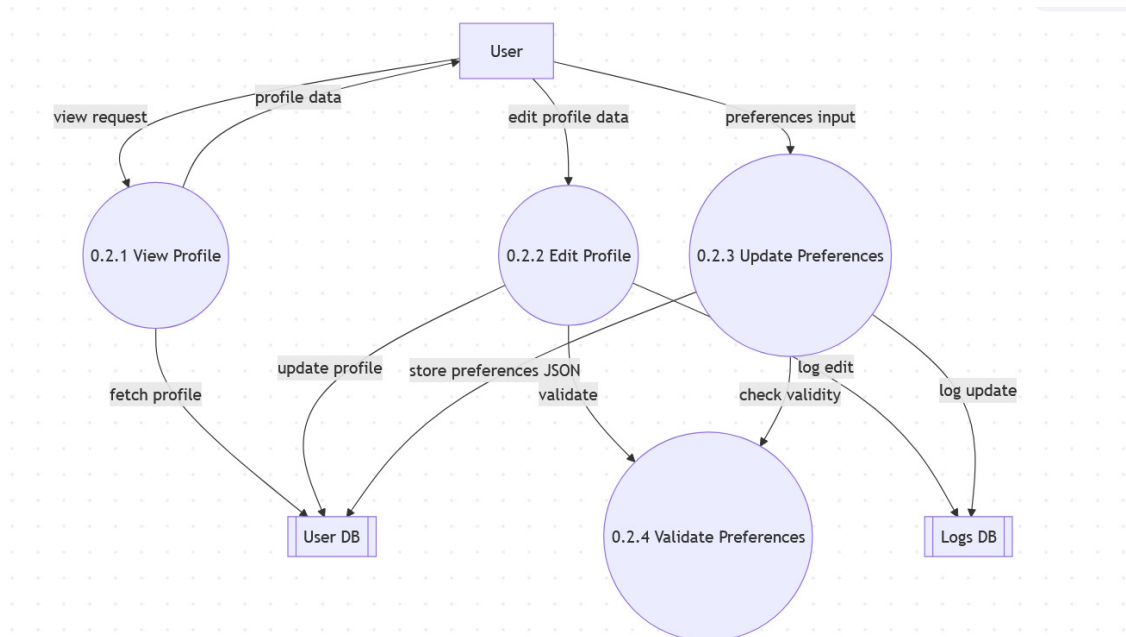


Figure 4: Level 2 DFD for 0.2 User Profile & Preferences

1.3.3 0.3 Task & List Management (CRUD)

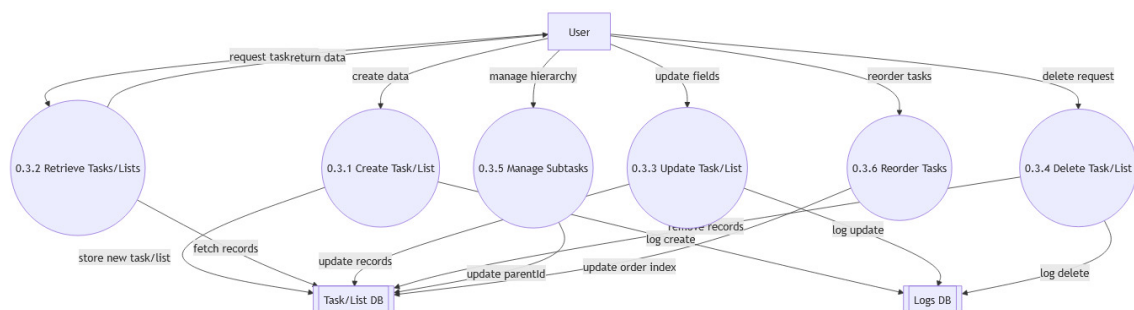


Figure 5: Level 2 DFD for 0.3 Task & List Management (CRUD)

1.3.4 0.4 Automated List Generation via LLM

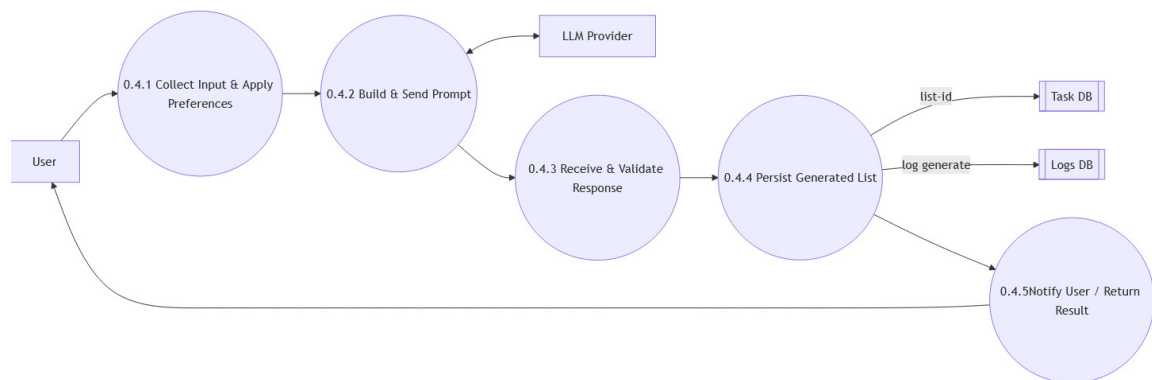


Figure 6: Level 2 DFD for 0.4 Automated List Generation via LLM

1.3.5 0.5 Task Editing & Completion

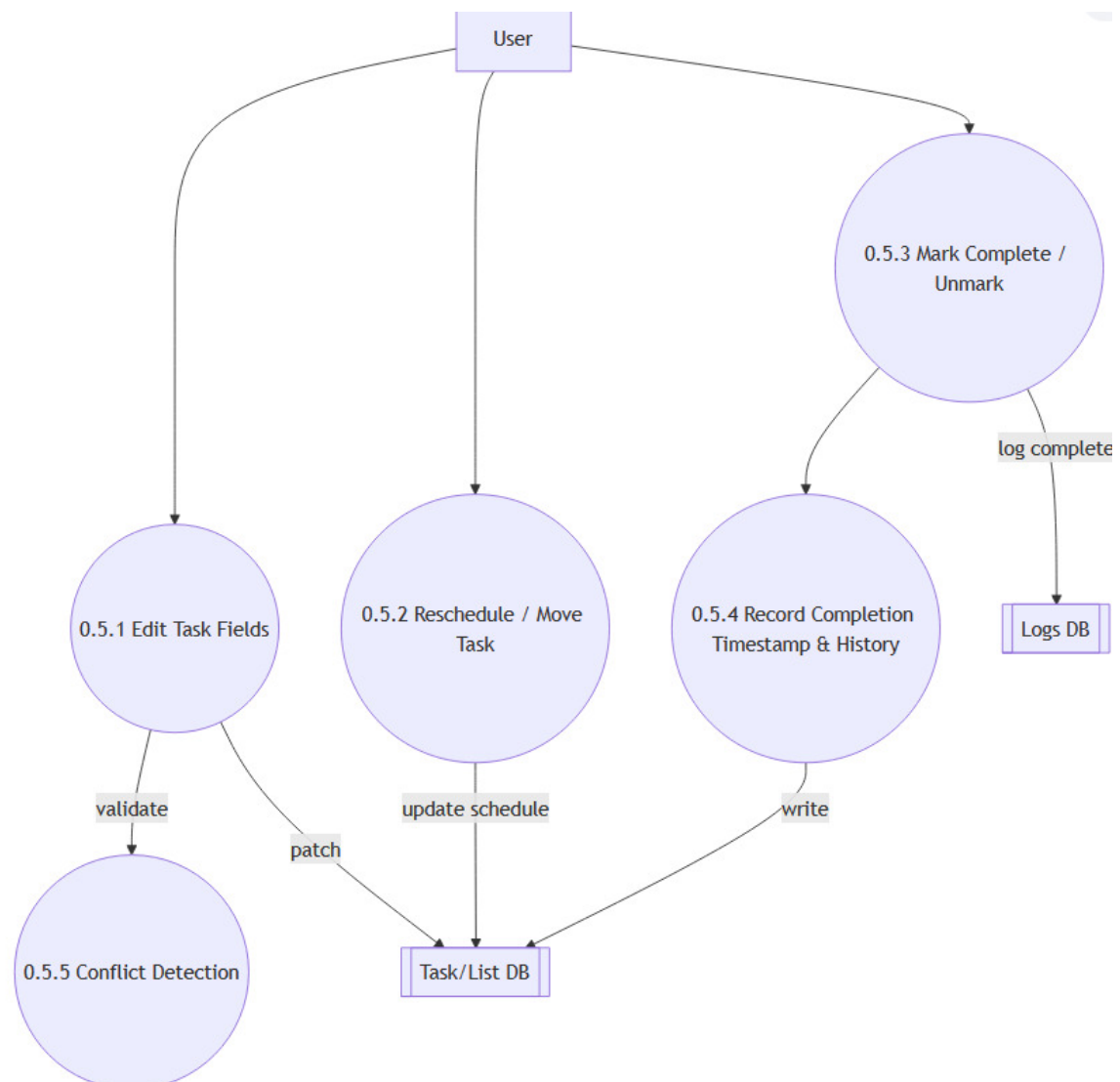


Figure 7: Level 2 DFD for 0.5 Task Editing & Completion

1.3.6 0.6 Exporting & Sharing

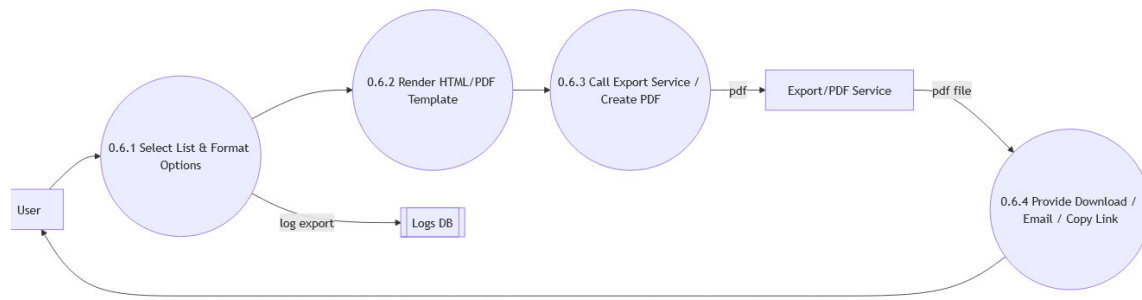


Figure 8: Level 2 DFD for 0.6 Exporting & Sharing

1.3.7 0.7 Logging & Audit



Figure 9: Level 2 DFD for 0.7 Logging & Audit

1.4 Data Dictionary

1.4.1 Terminology and Definitions

Term	Definition
Credentials	{email + password} provided by user for authentication.
Auth Token	Session token (JWT) issued after successful login.
User Profile	{userId, displayName, email, preferences, timestamps}.
Preferences	{workHours, timezone, dailyBuffer, generationLimits}.
Task	{id, listId, title, description, priority, scheduledAt, duration, tags, status}.
To-Do List	Collection of tasks linked to a user, possibly generated by LLM.
Generation Request	Free-text input + preferences sent to LLM.
LLM Response	Parsed JSON structure of generated tasks.
Export Request	Request to export tasks or lists into PDF/HTML.
Export File	Binary PDF returned to the user.
Log Entry	{id, userId, action, details, timestamp}.
Session Store	Active session records ensuring single-device login.
Share Token	Secure link for sharing exported or generated lists.

2 Core Modules

This section describes the main modules of the Automated To-Do List Application. Each module has been assigned to a designer for detailed implementation, and the functions under each module are listed.

2.1 0.1 User Authentication & Management

Handles user registration, login, OAuth integration, password reset, and session management.

Designer: Sanchit Srisha Udupa

Functions: Register User, Login using Email/Password, OAuth Callback & Link Accounts, Password Reset, Session Management

2.2 0.2 User Profile & Preferences

Manages user-specific settings such as working hours, reminders, and personalization.

Designer: Sanchit Srisha Udupa

Functions: View User Profile, Update Preferences, Apply Preferences to List Generation

2.3 0.3 Task & List Management (CRUD)

Enables creation, retrieval, updating, and deletion of tasks and lists.

Designer: Pranav Bhat P

Functions: Create Task/List, Retrieve Task/List, Update Task/List, Delete Task/List

2.4 0.4 Automated List Generation via LLM

Generates intelligent to-do lists based on user input and preferences using an LLM.

Designer: Pranav Bhat P

Functions: Send Generation Request, Receive and Parse LLM Response, Store Generated List

2.5 0.5 Task Editing & Completion

Allows modification of existing tasks and marking them as completed.

Designer: Pranav Bhat P

Functions: Edit Task Details, Mark Task as Completed

2.6 0.6 Exporting & Sharing

Provides functionality for exporting tasks/lists in different formats and sharing with others.

Designer: Pranav Vinodh

Functions: Export Task/List as PDF, Share via Secure Link

2.7 0.7 Logging & Audit

Maintains records of all critical actions for monitoring and security.

Designer: Pranav Vinodh

Functions: Log User Actions, Generate Audit Reports

3 Structure Charts

3.1 Level 1 Structure Chart

3.1.1 Module Decomposition

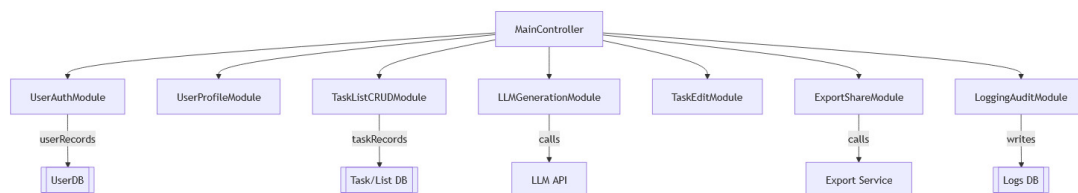


Figure 10: Level 1 Structure Chart of the System